QoS-QoE Translation with Large Language Model

Yingjie Yu, Mingyuan Wu, Ahmadreza Eslaminia, Lingzhi Zhao, Kaizhuo Yan, and Klara Nahrstedt
University of Illinois Urbana-Champaign, Urbana, Illinois, USA
{yyu69,mw34,ae15,lz26,kaizhuo2,klara}@illinois.edu

APPENDIX

A. Supervised Fine-tuning Instances

Figure 1 shows abridged chat-style supervised fine-tuning (SFT) instances for the two translation directions. Each instance is constructed from a single source-grounded relationship record. A system message instructs the model to output only the target field in valid JSON format. The user message contains the structured task input, including the task direction, contextual metadata, scenario, parameter mapping and definitions, source-grounded evidence, a partial history log, and a query. To create each training instance, we remove one time point from the original `history_log`. The observed side of that removed record is used as the `query`, and the corresponding target side is used as the assistant output. In the QoS→QoE setting, the query contains QoS value(s) and the assistant predicts the corresponding QoE value(s). In the QoE→QoS setting, the query contains QoE value(s) and the assistant predicts the corresponding QoS value(s).

B. History Log Construction

Many source papers report QoS–QoE relationships through mathematical formulations without releasing the underlying measurement data. To preserve these relationships for downstream translation tasks, we adopt an evidence-first strategy for constructing `history_log`. When numerical observations are available, we extract values from tables and text or approximate them from figures, supplementing them with synthetic samples when necessary. When no numerical observations are available, we construct synthetic samples guided by the extracted relationships and parameter definitions.

Each history log contains about 6–20 QoS–QoE observations. The resulting histories may therefore combine source-derived and simulated values or consist entirely of simulated values. While the extracted QoS–QoE relationships remain grounded in the source literature, the constructed histories should not be interpreted as independently measured observations from the original studies.

C. Train and Test Split and Leakage Audit

We split the dataset at the source record level before constructing supervised fine-tuning instances. All instances derived from the same source-grounded QoS–QoE relationship record are therefore assigned to the same split. The training split contains 913 source records and 7,205 instances, whereas the test split contains 113 source records and 902 instances.

We audit potential leakage using three overlap checks. Source record overlap checks whether any source-grounded relationship record contributes instances to both splits. Record task query overlap checks whether the same held-out query occurs under the same source record and translation task in both splits. Exact task query target overlap checks whether an identical normalized task, query, and target combination occurs across the two splits, including combinations derived from different source records. As shown in Table I, all three overlap counts are zero.

TABLE I: Leakage audit results for the train and test splits.

Overlap check	Count
Source record overlap	0
Record task query overlap	0
Exact task query target overlap	0

TABLE II: Canonical parameter-specific tolerance configuration used for Accuracy@δ for model evaluation. Here, *rel* and *abs* denote relative and absolute tolerance, respectively.

Parameter	General definition	Mode	δ
bitrate	transmission rate	rel	0.1
qoe_score	generic QoE score	abs	0.3
mos	subjective opinion score	abs	0.3
vmaf	video quality metric	abs	3.0
rebuffering_time	stall duration	abs	0.5
jitter	packet delay variation	abs	5.0
packet_loss_rate	packet loss ratio	abs	0.01
latency	end-to-end delay	abs	10.0
rebuffering_ratio	stall ratio	abs	0.03
bitrate_change	bitrate switching magnitude	rel	0.15
rebuffer_count	stall count	abs	1.0
psnr	peak signal fidelity	abs	1.5
visqol	perceptual quality metric	abs	0.3
ssim	structural similarity	abs	0.02
vifp	visual information fidelity	abs	0.03
vqm	video quality metric	abs	0.05
pesq	perceptual speech quality	abs	0.25
peaq_odg	audio difference grade	abs	0.2
playback_rate	playback speed	abs	0.05
frame_rate	frames per second	abs	2.0
snr	signal-to-noise ratio	abs	2.0
ssim_db	SSIM in decibels	abs	1.5
total_rebuffering_time	cumulative stall duration	abs	1.0
segment_duration	segment duration	abs	0.25
video_delay_p95_ms	p95 video delay	abs	20.0
acr_grade	absolute category rating	abs	0.5

D. Model Evaluation Details

For continuous-value prediction, we use MAPE and Accuracy@δ. All continuous values are normalized to canonical units before model evaluation to ensure consistent comparison across source papers. This section provides the parameter-



(a) QoS→QoE SFT instance.



(b) QoE→QoS SFT instance.

Fig. 1: Abridged chat-style SFT instances for bidirectional QoS-QoE translation. Long text fields and repeated entries are abbreviated with "..." for readability.

specific tolerance configuration for Accuracy@δ. We set the tolerance mode and δ value for each parameter based on its scale, unit, and semantic interpretation in the QoS-QoE translation setting. Because QoS and QoE parameters differ substantially in scale, units, and meaning, we use a parameter-specific δ rather than a single shared threshold. A prediction is counted as correct if its error falls within the parameter-specific tolerance, using absolute error for abs parameters and relative error for rel parameters. Table II reports the canonical parameter names, general definitions, and corresponding tolerance settings used in our evaluation. Notably, these general definitions are intended to provide a unified interpretation for evaluation, while the exact definition of the same parameter may vary slightly across different source papers.

Relative tolerances. Relative tolerances are used for scale-dependent rate-like quantities whose prediction error is more naturally interpreted proportionally than by a fixed absolute difference [1]. For `bitrate`, prior work also evaluates prediction quality using relative bitrate error, reflecting the fact that the same absolute error can correspond to very different levels of deviation across low and high bitrate ranges [2]. We use $\delta = 0.1$ for `bitrate`, which treats predictions within 10% of the ground truth as correct under Accuracy@δ. This provides a moderate proportional tolerance for near-correct predictions while preserving separation between materially different bitrate values. We use the same tolerance mode for `bitrate_change`, with $\delta = 0.15$.

Absolute tolerances. Absolute tolerances are used when prediction error is more naturally interpreted on the original scale, making deviations directly meaningful in the native unit or rating scale [1], [3]. In our setup, we use this tolerance mode for bounded quality scores, objective quality metrics, time-based metrics, ratios, counts, and delay-related metrics.

Bounded subjective and perceptual quality scores are defined on fixed rating scales, so their prediction error is more naturally interpreted on the original scale than as proportional error [4], [5]. This category includes `qoe_score` ($\delta = 0.3$), `mos` ($\delta = 0.3$), `visqol` ($\delta = 0.3$), `pesq` ($\delta = 0.25$), `peaq_odg` ($\delta = 0.2$), and `acr_grade` ($\delta = 0.5$). We set these δ values as small absolute tolerances on their scales to allow minor deviations while preserving meaningful quality differences.

Objective fidelity metrics are commonly evaluated on their own metric-specific scales [6], [7]. This category includes `vmaf` ($\delta = 3.0$), `psnr` ($\delta = 1.5$), `ssim` ($\delta = 0.02$), `vifp` ($\delta = 0.03$), and `vqm` ($\delta = 0.05$). We set these δ values as small absolute tolerances relative to the range and resolution of each metric.

Signal and native-unit rate metrics are expressed in engineering units such as decibels or frames per second [1], [8]. This category includes `snr` ($\delta = 2.0$), `ssim_db` ($\delta = 1.5$), and `frame_rate` ($\delta = 2.0$). We set these δ values as small absolute tolerances in their native units, allowing modest deviations while preserving materially different signal or temporal characteristics.

Time-based and delay-related metrics are measured in units such as seconds or milliseconds [1], [9]. This category includes `rebuffering_time` ($\delta = 0.5$), `total_rebuffering_time` ($\delta = 1.0$), `latency` ($\delta = 10.0$), `jitter` ($\delta = 5.0$), `segment_duration` ($\delta = 0.25$), `video_delay_p95_ms` ($\delta = 20.0$), and `playback_rate` ($\delta = 0.05$). We set these δ values as small absolute tolerances in their native units to allow near-correct timing predictions without conflating materially different delay behavior.

Ratio- and count-based parameters are interpreted on their original scales because they represent bounded fractions

TABLE III: Average rating and confidence of each LLM reviewer during the first-round evaluation.

LLM reviewer	Avg. rating	Avg. confidence
Gemini-2.5-flash-lite	8.48	5.00
Claude-haiku-4-5-20251001	7.24	4.08
Grok-4.20-0309-reasoning	6.20	3.98

TABLE IV: Binary reject versus non-reject agreement among the LLM reviewers over all 1,026 evaluated records.

Reviewer set	Agreement (%)
All three reviewers	73.49
(Claude-haiku-4-5-20251001, Gemini-2.5-flash-lite)	81.29
(Claude-haiku-4-5-20251001, Grok-4.20-0309-reasoning)	86.75
(Gemini-2.5-flash-lite, Grok-4.20-0309-reasoning)	78.95

or discrete events rather than continuously scaled magnitudes [1], [10]. This category includes `packet_loss_rate` ($\delta = 0.01$), `rebuffering_ratio` ($\delta = 0.03$), and `rebuffer_count` ($\delta = 1.0$). We set these δ values as small absolute tolerances, allowing minor deviations for bounded ratios and a one-event tolerance for counts.

E. Multi-LLM Reviewer Evaluation Details

To improve data quality, each metadata-enriched record is evaluated by three LLM reviewers: Gemini-2.5-flash-lite [11], Claude-haiku-4-5-20251001 [12], and Grok-4.20-0309-reasoning [13]. Each reviewer returns a rating, a confidence score, and textual feedback. Table III summarizes the average rating and confidence of each reviewer during the first-round evaluation, providing a coarse view of reviewer behavior and calibration. Gemini-2.5-flash-lite assigns the highest average rating and confidence, while Grok-4.20-0309-reasoning assigns the lowest. Claude-haiku-4-5-20251001 lies between the other two reviewers on both measures. These differences indicate that the reviewers are not identically calibrated and motivate examining their agreement on the pipeline's primary filtering decision.

Reviewer agreement. We further evaluate the consistency of the three LLM reviewers over all 1,026 records using their binary reject versus non-reject decisions. The binary decisions are derived using the predefined filtering rule employed by the data evaluation pipeline. The three reviewers assign the same binary decision to 73.49% of the records. Pairwise agreement is 81.29% between Claude-haiku-4-5-20251001 and Gemini-2.5-flash-lite, 86.75% between Claude-haiku-4-5-20251001 and Grok-4.20-0309-reasoning, and 78.95% between Gemini-2.5-flash-lite and Grok-4.20-0309-reasoning. As summarized in Table IV, these results show generally high agreement on the primary filtering decision while also revealing reviewer-specific differences. The multi-reviewer design limits dependence on the calibration and judgments of any single model.

F. Human Audit Details

We conduct two-annotator audits at the paper curation, re-evaluation, and final dataset stages. At each stage, two annotators independently assess 50 samples drawn uniformly at random without replacement using stage-specific criteria: candidate source papers, paired LLM reviewer comments and resulting re-evaluated records, and records from the final dataset, respectively. Both annotators had prior research experience in multimedia systems and QoS–QoE analysis.

Metrics. Across all human audits, we report three complementary metrics. A positive overall judgment denotes validity for candidate papers and records and usefulness for LLM reviewer comments. The unanimous positive rate is the number of sampled items receiving a positive overall judgment from both annotators divided by the total number of sampled items. The criterion-level pass rate is the number of pass judgments divided by the total number of applicable criterion judgments made by both annotators across all sampled items. The agreement rate is the number of sampled items for which the two annotators assign the same overall judgment divided by the total number of sampled items, including both positive and negative agreement. Together, these metrics capture joint positive assessment, criterion-level quality, and inter-annotator consistency.

Paper curation audit. The curation audit evaluates 50 candidate papers using four inclusion criteria:

- **P1: Multimedia delivery relevance.** The paper studies a multimedia delivery, streaming, communication, or system setting.
- **P2: QoS–QoE relationship presence.** The paper contains an explicit or recoverable relationship between at least one QoS factor and one QoE outcome.
- **P3: Extractable evidence.** The relationship is supported by extractable evidence such as text, a table, a figure, an equation, or reported measurements.
- **P4: Sufficient contextual metadata.** The paper provides enough contextual information to interpret and normalize the relationship.

Together, P1 and P2 assess whether a candidate paper is relevant to the dataset scope and contains a usable QoS–QoE relationship, while P3 and P4 assess whether the relationship can be extracted, verified, and normalized with sufficient source context. As shown in Table V, the audit achieves a 94.0% unanimous positive rate, a 98.0% criterion-level pass rate, and a 94.0% agreement rate. These results indicate that nearly all individual criterion judgments are positive and that the two annotators generally agree on the overall inclusion decision. Specifically, 47 of the 50 sampled papers receive positive overall judgments from both annotators, while the annotators disagree on the remaining three.

Re-evaluation audit. The re-evaluation audit examines 50 cases sampled from the re-evaluation stage. Each case contains the LLM reviewer comments and the resulting re-evaluated record. The reviewer comments are assessed using three criteria:

TABLE V: Two-annotator human audit results (%).

Audit target	Unanimous positive rate	Criterion-level pass rate	Agreement rate
Candidate papers	94.0	98.0	94.0
LLM reviewer comments	80.0	95.3	80.0
Re-evaluated records	72.0	96.4	78.0
Final dataset records	74.0	91.6	78.0

- **R1: Specific issue identification.** The comment identifies a concrete issue in the original record.
- **R2: Issue validity.** The identified issue is supported by the source evidence or record content.
- **R3: Actionability.** The comment provides information that can guide a meaningful correction.

The re-evaluated records are assessed using five correctness criteria:

- **C1: QoS correctness.** The QoS parameters, values, and definitions are correctly represented.
- **C2: QoE correctness.** The QoE parameters, values, and definitions are correctly represented.
- **C3: Relationship correctness.** The extracted relationship, value, equation, and direction are correct.
- **C4: Source-evidence support.** The record is supported by the cited source evidence.
- **C5: Metadata consistency.** The contextual metadata is consistent with the source paper and extracted relationship.

Together, R1–R3 assess whether the LLM reviewer comments identify concrete and valid problems and provide actionable guidance, while C1–C5 assess whether the resulting records are correct, supported by source evidence, and contextually consistent. As summarized in Table V, the LLM reviewer comments achieve an 80.0% unanimous positive rate, a 95.3% criterion-level pass rate, and an 80.0% agreement rate. The corresponding rates for the re-evaluated records are 72.0%, 96.4%, and 78.0%. The high criterion-level pass rates indicate that most individual criterion judgments are positive, whereas the lower unanimous positive rates reflect the stricter requirement that both annotators assign a positive overall judgment to each comment or record.

Final dataset audit. The final dataset audit evaluates an independent sample of 50 records drawn from the full dataset using the same C1–C5 record correctness criteria defined above. It therefore estimates the quality of the final dataset rather than only that of records routed through the re-evaluation process. As summarized in Table V, the audit achieves a 74.0% unanimous positive rate, a 91.6% criterion-level pass rate, and a 78.0% agreement rate. Among the individual criteria, source evidence support achieves a 100% pass rate, and metadata consistency achieves a 96% pass rate. The high criterion-level pass rate indicates that most individual criterion judgments are positive, while the lower unanimous positive rate reflects the stricter requirement that both annotators assign a positive overall judgment to each record as a whole.

Confidence Intervals. We compute 95% Wilson score confidence intervals for the unanimous positive and agreement rates using 50 audited items per stage. For candidate papers, re-evaluated records, and final dataset records, the confidence intervals for unanimous positive rates are [83.8%, 97.9%], [58.3%, 82.5%], and [60.4%, 84.1%], respectively. The corresponding intervals for agreement rates are [83.8%, 97.9%], [64.8%, 87.2%], and [64.8%, 87.2%].

Error Analysis. We further analyze extraction quality across source-evidence types using the 50 audited final dataset records. Among 20 equation-based, 26 figure-based, and 4 table-based records, relationship correctness (C3) is flagged by at least one annotator for 5 (25.0%), 3 (11.5%), and 0 (0%) records, respectively. None of these records is marked incorrect by both annotators, indicating annotation disagreements rather than jointly confirmed extraction errors.

At the record level, at least one annotator identifies validity concerns in 8 of 20 (40.0%) equation-based records and 5 of 26 (19.2%) figure-based records. Within equation-based records, mathematical relationships involving multiple QoS input parameters exhibit a higher rate of overall validity concerns than those involving a single QoS input (45.5% versus 33.3%). However, their respective C3 flag rates are 18.2% and 33.3%, showing that the higher overall concern rate does not translate into more relationship correctness flags. These findings identify equation-based relationships, particularly multi-variable formulations, as a recurring area of record-level quality concerns in the audited sample.

At the source-paper level, QoE parameter classification (C2) is flagged in 9 of the 50 records (18.0%). These include relationships involving objective performance indicators, such as frame rate, video resolution, bitrate switching rate, and buffering delay, where annotators disagree on the distinction between QoE outcomes and system-performance metrics. Such classification disagreements represent another recurring category of correctness concerns in the audited records.

G. Pipeline Prompt Details

The full prompts used in our pipeline are included in the anonymized supplementary material. In particular, `relationship_extraction.txt` is used for source-grounded QoS–QoE relationship extraction from parsed papers, `metadata_enrichment.txt` is used for contextual metadata generation and normalization, and `data_evaluation.txt` is used in the multi-reviewer data evaluation stage. Together, these prompts define the task-specific instructions and structured output requirements used in the dataset construction pipeline.

REFERENCES

[1] J. Terven, D.-M. Cordova-Esparza, J.-A. Romero-González, A. Ramírez-Pedraza, and E. A. Chávez-Urbiola, "A comprehensive survey of loss functions and metrics in deep learning," *Artif. Intell. Rev.*, vol. 58, 2025, art. no. 195.

[2] T. Sharma, T. Mangla, A. Gupta, J. Jiang, and N. Feamster, "Estimating WebRTC video QoE metrics without using application headers," in *Proc. ACM Internet Meas. Conf. (IMC)*, 2023. [Online]. Available: https://doi.org/10.1145/3618257.3624828

[3] S. Beddar-Wiesing, A. Moallemy-Oureh, M. Kempkes, and J. M. Thomas, "Absolute evaluation measures for machine learning: A survey," *arXiv:2507.03392*, 2025.
[4] International Telecommunication Union, "Methods for subjective determination of transmission quality," ITU-T, Recommendation P.800, Aug. 1996. [Online]. Available: https://www.itu.int/rec/T-REC-P.800-199608-I
[5] ——, "Application guide for objective quality measurement based on recommendations p.862, p.862.1 and p.862.2," ITU-T, Recommendation P.862.3, Nov. 2005. [Online]. Available: https://www.itu.int/rec/T-REC-P.862.3-200511-S
[6] A. K. Venkataramanan, C. Stejerean, I. Katsavounidis, and A. C. Bovik, "One transform to compute them all: Efficient fusion-based full-reference video quality assessment," *IEEE Trans. Image Process.*, vol. 33, pp. 509–524, 2024.
[7] M. H. Pinson and S. Wolf, "A new standardized method for objectively measuring video quality," *IEEE Trans. Broadcast.*, vol. 50, no. 3, pp. 312–322, 2004.
[8] A. MacKin, A. Zhang, and D. R. Bull, "A study of subjective video quality at various frame rates," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, 2015, pp. 3407–3411.
[9] J. J. Aranda, M. Cortés, J. Salvachúa, M. Narganes, and I. Martínez-Sarriegui, "The quality for service (Q4S) protocol," RFC Editor, RFC 8802, Jul. 2020. [Online]. Available: https://www.rfc-editor.org/rfc/rfc8802.html
[10] F. Dobrian *et al.*, "Understanding the impact of video quality on user engagement," *ACM SIGCOMM Comput. Commun. Rev.*, vol. 41, no. 4, pp. 362–373, 2011.
[11] Google, "Gemini 2.5 flash-lite," 2025, accessed: 2026-04-01. [Online]. Available: https://ai.google.dev/gemini-api/docs/models/gemini-2.5-flash-lite
[12] Anthropic, "Introducing claude haiku 4.5," Oct. 2025, accessed: 2026-04-01. [Online]. Available: https://www.anthropic.com/news/claude-haiku-4-5
[13] xAI, "Grok 4.20 0309 reasoning," accessed: 2026-04-01. [Online]. Available: https://docs.x.ai/developers/models/grok-4.20-0309-reasoning